Tien-Chen Lin

Cytoskeletal neurobiology · Mechanobiology of brain development

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Research summary

How do cytoskeletal systems turn transient dynamics into stable cellular form? My research has addressed this question across scales: from biochemically resolving how the microtubule-nucleation machinery is activated in space and time, to discovering in my first-author *Nature* study that competing actin networks form an intrinsic oscillator, a mechanical system of local activation and global inhibition that establishes neuronal polarity. My independent program asks how physiological neuron–glia interactions move neurons between cytoskeletal states, and how transient cytoskeletal decisions are consolidated into persistent neural architecture. I combine calibrated local perfusion of single cells, quantitative live imaging, mechanistic biochemistry and dynamical-systems modeling to identify control parameters governing how neural form is established, maintained and reset.

Education

2009–2014 **Ph.D., Natural Sciences (Dr. rer. nat., *magna cum laude*)**, *University of Heidelberg*, Germany
Defended 12 November 2014. ZMBH, laboratory of Prof. E. Schiebel.

2005–2006 **M.Sc., Biochemistry & Molecular Biology**, *National Yang-Ming University (now National Yang Ming Chiao Tung University)*, Taiwan
Awarded through the 5-year combined B.Sc./M.Sc. fast-track program.

2002–2005 **B.Sc., Life Sciences**, *National Yang-Ming University (now National Yang Ming Chiao Tung University)*, Taiwan

Research experience

2017–present **Postdoctoral Researcher**, *DZNE, Bonn — Axon Growth & Regeneration (Prof. F. Bradke)*
Conceived the project (with C.H. Coles and F. Bradke) and led the study showing that a soma-based ARP2/3–myosin-II local-activation, global-inhibition system establishes CNS neuronal polarity; reconciled the roles of actin networks in neurite growth; built an R/ImageJ analysis pipeline for neurite-growth dynamics; collaborative omics of axon regeneration, reactive astrocytes after stroke, and DRG-neuron subtypes.
Position funded by [ADD: source]; current contract to [ADD: date].

2014–2017 **Postdoctoral Researcher**, *ZMBH, University of Heidelberg (Prof. E. Schiebel)*
Established MOZART1 as an integral part of the γ-tubulin small complex in *Candida albicans* and human cells, and as a driver of its assembly into a nucleation-competent ring; mapped conserved interactions among MOZART1, γ-TuSC, and the receptors pericentrin and CDK5RAP2.

2009–2014 **Doctoral Researcher**, *ZMBH, University of Heidelberg (Prof. E. Schiebel)*
Elucidated the mitotic phospho-regulation of the γ-tubulin small complex and pericentrin; characterized γ-TuSC–receptor interactions and the divergent targeting of γ-tubulin complexes to microtubule organizing centers.

2007–2009 **Research Assistant**, *Genomics Research Center, Academia Sinica, Taipei (Prof. Ming-Daw Tsai)*, Taiwan
Structural characterization of HICE1, HEC1 and HACE1; biochemical characterization of the GRP78–NPGPx interaction.

2006–2007 **Compulsory military service**, *Taiwan*

Fellowships & awards

2016 **EMBO Short-term Fellowship**
Awarded for the *Candida albicans* γ-tubulin small complex / MOZART1 project; supported a collaboration with Prof. R. Arkowitz, Université Côte d'Azur.

2012 **LGFG Fellowship (Landesgraduiertenförderung) — "Regulation of Cell Division"**
[ADD: any further grants, travel awards or prizes, 2017–2026.]

Publications

Full list: tien-chen-lin.github.io/publications · ORCID 0000-0002-6852-2900

Metrics ~730 citations across nine indexed papers (mean ~81 per paper; **8 of 9 cited more than 60 times**). *(Google Scholar, July 2026; the 2026* Nature *paper is not yet indexed for citations.)*

Key papers *Nature* 2026: whole-cell competition between actin networks selects the axon. *J. Cell Biol.* 2016: reconstitution of an active microtubule-nucleation template. *eLife* 2014: phosphorylation licenses nucleation in space and time. Together, molecular activation → assembly-machine reconstitution → whole-cell self-organization.

Research

2026 **Lin, T.-C.**, Coles, C.H., Alfadil, E., Fäßler, F., Husch, A., Dupraz, S., Pietralla, T., Narita, A., Schelski, M., Flynn, K.C., Stern, S., Möhl, C., Hilton, B.J., Vauti, F., Arnold, H.-H., Schur, F.K.M., Bradke, F. An intrinsic cytoskeletal oscillator establishes neuronal polarity. *Nature* (2026). doi:10.1038/s41586-026-10755-6

2016 **Lin, T.-C.**, Neuner, A., Flemming, D., *et al.* MOZART1 and γ-tubulin complex receptors are both required to turn γ-TuSC into an active microtubule nucleation template. *J. Cell Biol.* 215(6), 823–840. *(65 citations)*

2014 **Lin, T.-C.**, Neuner, A., Schlosser, Y.T., *et al.* Cell-cycle dependent phosphorylation of yeast pericentrin regulates γ-TuSC-mediated microtubule nucleation. *eLife* 3, e02208. *(106 citations)*

2011 **Lin, T.-C.**, Gombos, L., Neuner, A., *et al.* Phosphorylation of the yeast γ-tubulin Tub4 regulates microtubule function. *PLoS ONE* 6(5), e19700. *(64 citations)*

Review

2015 **Lin, T.-C.**[#], Neuner, A., Schiebel, E.[#] Targeting of γ-tubulin complexes to microtubule organizing centers: conservation and divergence. *Trends Cell Biol.* 25(5), 296–307. [#]*Corresponding author. (150 citations — my most-cited work.)*

Co-authored

2022 Kugler, C., Blank, N., Matuskova, H., *et al.*, **Lin, T.-C.**, Bradke, F., Petzold, G.C. Pregabalin improves axon regeneration and motor outcome in a rodent stroke model. *Brain Commun. Contribution: transcriptomic re-analysis identifying extracellular-matrix gene upregulation in reactive astrocytes.*

2022 Hilton, B.J., Husch, A., Schaffran, B., **Lin, T.-C.**, *et al.* An active vesicle priming machinery suppresses axon regeneration upon adult CNS injury. *Neuron* 110, 51–69.e7. *(107 citations) Contribution: transcriptomic re-analysis linking presynaptic vesicle gene downregulation to regenerative capacity.*

2018 Gunzelmann, J., Rüthnick, D., **Lin, T.-C.**, *et al.*, Schiebel, E. The microtubule polymerase Stu2 promotes oligomerization of the γ-TuSC for cytoplasmic microtubule nucleation. *eLife* 7, e39932. *(80 citations)*

2015 Chinen, T., Liu, P., Shioda, S., *et al.*, **Lin, T.-C.**, *et al.* The γ-tubulin-specific inhibitor gatastatin reveals temporal requirements of microtubule nucleation during the cell cycle. *Nat. Commun.* 6, 8722. *(74 citations)*

2014 Elserafy, M., Šarić, M., Neuner, A., **Lin, T.-C.**, *et al.*, Schiebel, E. Molecular mechanisms that restrict yeast centrosome duplication to one event per cell cycle. *Curr. Biol.* 24, 1456–1466. *(75 citations)*

In preparation

Modeling of cytoskeletal oscillation dynamics in neuronal polarization (research article) — **Lin, T.-C.** (first and corresponding author).

Cytoskeletal self-organization in neurons (review article) — **Lin, T.-C.** (co-corresponding author).

Local perfusion for acute manipulation of single neurons (protocol article) — **Lin, T.-C.** (corresponding author).

Technical expertise

Biochemistry Recombinant protein expression and purification; *in vitro* reconstitution of multi-subunit complexes; single-particle electron microscopy; electron tomography.

Imaging Live-cell, TIRF and super-resolution microscopy; tissue clearing; quantitative image analysis and automated segmentation.

Perturbation Optogenetic and chemogenetic control of cytoskeletal proteins; mouse genetics; *in utero* electroporation; organotypic and cortical slice culture; local perfusion of single neurons.

Computational R, Python and ImageJ macro development; custom pipelines for neurite-growth dynamics; bulk and single-cell transcriptomic analysis; dynamical-systems modeling (in collaboration).

Supervision & teaching

2023–2025 **Module coordinator**, *M.Sc. "Molecular Cell Biology" (EM5 practical module), University of Bonn*
Designed the curriculum and schedule, managed technical and human resources, liaised with the program office, and supervised teaching within the module.

2023 **Lecturer**, *M.Sc. "Molecular Cell Biology" (EM5), University of Bonn*
Data Visualization with R (4 h) — principles of graphical communication and `ggplot2` practice.

2020, 2022 **Lecturer**, *M.Sc. "Molecular Cell Biology" (EM5), University of Bonn*
Light Microscopy Basics (2 h each) — optics, objective selection, experimental design, and current techniques.

2022 **Instructor**, *M.Sc. "Neuroscience", University of Bonn*
Live-cell imaging practical (2 days) — neuronal electroporation, imaging of cytoskeletal dynamics, ImageJ analysis and macro scripting.

2013 **Instructor**, *M.Sc. practical "Mechanism and Regulation of Eukaryotic Cell Division", University of Heidelberg*
One-week course on budding yeast as a model for cell-cycle checkpoints; protein-complex isolation and microtubule assembly.

2012 **Lecturer**, *M.Sc. "Cell Cycle Checkpoint Regulation", University of Heidelberg*
(2 h)

2025 **Research supervision — M.Sc. rotation**, *DZNE Bonn*
Josephine Hahn (M.Sc. Neuroscience), one-month laboratory rotation: *Influence of ARP2/3 and myosin II on microtubule retrograde flow in developing hippocampal neurons.* Designed and supervised a complete project — live imaging of CAMSAP3/LifeAct in primary hippocampal neurons, acute pharmacological perturbation (CK-666, para-nitroblebbistatin, combined), kymograph analysis in Fiji and quantification in R — yielding an additive slowdown of microtubule retrograde flow under combined ARP2/3 and myosin II inhibition across four independent experiments.

2017 **Research supervision — M.Sc. project**, *ZMBH, University of Heidelberg*
Annalena Meyer: investigation of the GRIP1 domain of the γ-tubulin complex subunit GCP3.

2014 **Research supervision — M.Sc. project**, *ZMBH, University of Heidelberg*
Yvonne Schlosser: investigation of the N-terminal domain of the γ-tubulin complex subunit GCP3. Co-author on Lin *et al.*, *eLife* 2014.
[ADD: any further M.Sc./rotation students supervised at DZNE Bonn 2017–2026, and any co-supervision of doctoral researchers — a search committee reads supervision history as evidence you can run a group.]

Talks (selected)

2024–2026 **[ADD — the record currently stops at 2023; add anything since, especially talks given after the *Nature* paper appeared.]**

2023 **Microtubules in Neurons, Chiemsee; The Cytoskeleton of Neurons and Glia webinar series (virtual)**

2022 **CSHL meeting "Molecular Mechanisms of Neuronal Connectivity", New York; FENS Symposium "Barriers and hopes for functional restoration after traumatic CNS injury", Paris; EMBO Workshop "Neural development and neurodegeneration", Taipei; European Cytoskeleton Forum, Hannover; DZNE Mini Symposium "Information Flow", Bonn; Axon Repair Meeting (virtual)**

2022 **Talks in Taiwan: International Center for Wound Repair and Regeneration, College of Medicine, National Cheng Kung University, Tainan; Department of Biological Science and Technology, National Yang Ming Chiao Tung University, Hsinchu; lecture with F. Bradke, College of Life Sciences, National Yang Ming Chiao Tung University**

2021 **DZNE Science Meeting, Bonn**

2016 **DKFZ–ZMBH Alliance Young Investigators Seminar, Heidelberg**

Professional service

2019–present Associate Faculty Member, H1 Connect (formerly F1000 / Faculty Opinions) — peer recommendations of published work in axon regeneration and membrane mechanics.

Peer review [ADD: journals reviewed for.]

[ADD: conference or session organization, committee service, society memberships.]

Professional development

2021 Project Management for Scientists, EMBO.

2020 Laboratory Leadership for Group Leaders, EMBO; Negotiation for Scientists, EMBO; Advanced Techniques in Molecular Neuroscience, Cold Spring Harbor Laboratory.

Languages

[ADD: Mandarin Chinese, English, German — with levels.]

Referees

Prof. Frank Bradke — DZNE Bonn, Germany. [ADD: email]
Prof. Elmar Schiebel — ZMBH, University of Heidelberg, Germany. [ADD: email]
[Third referee — see the referee memo.] [ADD: affiliation, email]
Letters will be sent directly to the Institute, as requested.